TAKUDZWA WILSON MUZAVAZI
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GIS professional specializing in spatial analysis, remote sensing, geospatial data management, and environmental monitoring. Proficient in ArcGIS Pro, QGIS, Python, and R, experienced in analyzing complex datasets, managing geodatabases, processing satellite imagery, and producing accurate maps to support environmental and infrastructure projects.

EDUCATION

Master of Science in Geographic Information Science *Expected April 2027*
Michigan Technological University
Bachelor of Science in Geographic Information Science and Earth Observation *December 2020*
University of Zimbabwe

TECHNICAL SKILLS

ArcGIS Pro, QGIS | Spatial Analysis, Geoprocessing, ModelBuilder, Hotspot Analysis, Raster and Vector Analysis | Remote Sensing | Geodatabase Management, Data Validation | Cartography | R, Python | Field Data Collection (GPS, ArcGIS Field Maps, Open Data Kit)

PROFESSIONAL EXPERIENCE

Research Assistant, Michigan Technological University, Houghton, Michigan *August 2025 – Present*
- Analyze the spatial distribution of moose skeletal remains using ArcGIS Pro to investigate ecological patterns and habitat-related mortality.
- Maintain a geodatabase of more than 5,000 moose records, integrating skeletal records, GPS locations, and ecological attributes.
- Create maps and spatial visualizations using ArcGIS Pro and R to communicate wildlife distribution patterns and support ecological research.

GIS Assistant, Map Tech Pvt Ltd t/a Zimgeo, Harare, Zimbabwe *May 2024 – May 2025*
- Conducted GIS analysis of agricultural land use and mining environmental data to support environmental monitoring and planning.
- Georeferenced and digitized historical aerial photographs in QGIS for integration with modern spatial datasets.
- Processed Landsat and Sentinel-2 imagery in QGIS to analyze land-use change and environmental trends, supporting environmental planning and decision-making

Environmental Assistant, Glenrise, Ngezi, Zimbabwe *February 2022 – September 2022*
- Analyzed 125 mining incident records in QGIS using buffering and kernel density analysis to identify operational risk zones and support mine safety planning.
- Compiled geospatial data, field observations, and statistical summaries for environmental monitoring and compliance reports.
- Produced thematic maps and spatial reports to communicate environmental findings to project stakeholders.

GIS Intern, National Railways of Zimbabwe, Harare, Zimbabwe *June 2018 – July 2019*
- Collected and validated railway infrastructure and derailment-location data using GPS and satellite imagery, integrating information into GIS layers for analysis.
- Applied hotspot and spatial pattern analysis in QGIS to identify derailment risk zones and support targeted maintenance planning.
- Digitized and mapped 600 km of railway infrastructure for network analysis, infrastructure planning, and operational risk assessment.

PROFESSIONAL AFFILIATIONS AND LEADERSHIP

GIS Student Volunteer, GIS-Pro 2026 | Milwaukee, Wisconsin *October 2026*
- Supported workshop delivery, hands-on activities, and certification administration for the Cartography and Map Design Boot Camp: Theory and Best Practice Fundamentals.

Member | Geospatial Professional Network (GPN) *April 2026 – Present*

Member | American Association of Geographers (AAG) *April 2026 – Present*

Secretary of African Student Organization (ASO), Michigan Technological University *April 2026 – Present*
- Assist in the planning and execution of cultural events and community outreach programs promoting African heritage on campus.